

EMBEDDINGS MEMO

To: Professor Low

From: *Team 2*

Subject: Preferred Embedding Model for Text Queries

Input Text:

We want to use the summary text for our analysis because both the silhouette score and the yfinance analysis indicate that it produces the best overall results. We are confident that the summary provides rich information, offers sufficient context for our embedding models, and effectively distinguishes between different categories of companies. Additionally, we intend to include the extra VCard data, as it enhances the initial summary vector and improves the results across the table.

Model:

Based on the results, we have determined that the nomic-ai model is the most effective for this embedding task, as it achieves the highest classification scores in terms of silhouette and precision. Although it does experience challenges with attentiveness when processing large amounts of context data, we found that reducing text length through chunking or summarization yields the best results.

Nomic Prefix:

We have discovered that, regardless of the task, classification performs the best when input into the nomic-ai model. Therefore, we have decided to use that prefix for the remainder of our project. This is likely because the model can concentrate on creating classification buckets for organizing companies, allowing it to embed the text based on that focus rather than for simple searching. As a result, it produces the most meaningful information.